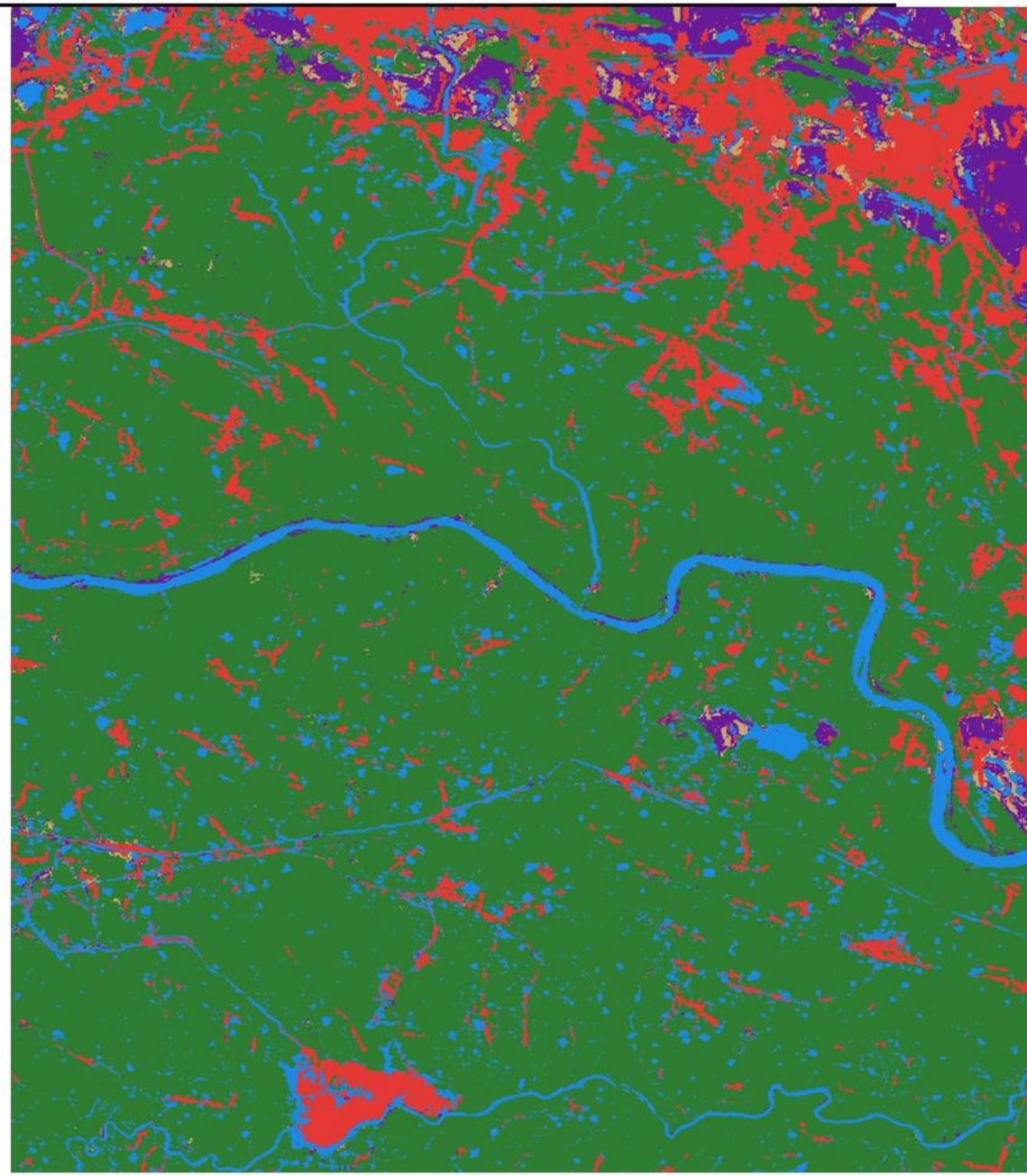
USER MANUAL · WALKTHROUGH

Custom LULC

Map any part of India at 10 m, in the classes you need. A short manual, and the index for the walkthrough video.

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Guided by Prof. Aaditeshwar Seth
M.Tech CSE
IIT Delhi
13 October 2026



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USING IT

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What it does

START

Sign in with Google
Name the project
Area: preset, coordinates or a box
A year
The base classes to start from

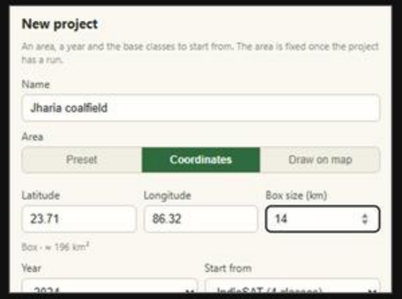
REFINE

Pick a class to split
Use a model from the zoo
or upload your own polygons
Train, read the held-out scores
Run again; every run is kept

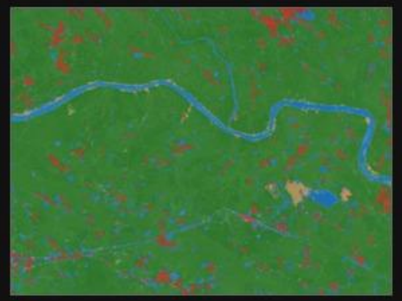
SHARE

Download the GeoTIFF
Download the whole project
A STAC record per run
Make the project public
Your model helps the next user

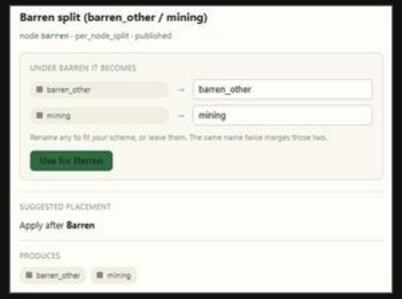
THE FIVE STEPS



1 **Create a project**: an area, a year and the base classes



2 **Run classification**: a 10 m base map of greenery, water, built-up and barren



3 **Split a class**: a model from the zoo, or a few polygons of your own



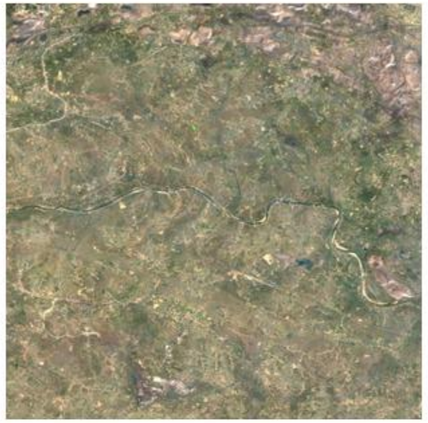
4 **Run again**: every run is kept with the scheme that made it



5 **Download and share**: GeoTIFF, project zip, STAC record, or make it public

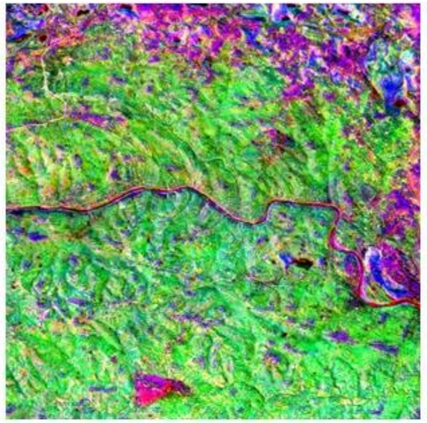
How it works

WHAT HAPPENS ON RUN



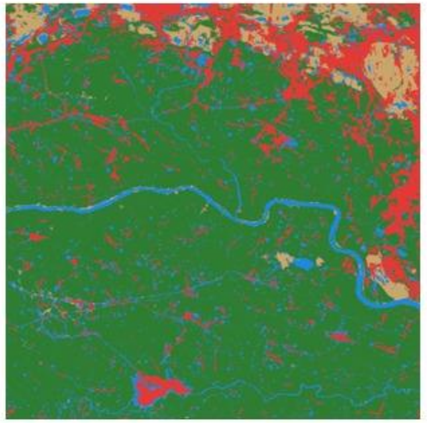
1 Your area and year go to Earth Engine

Jharia coalfield, 2024 (satellite view for reference)

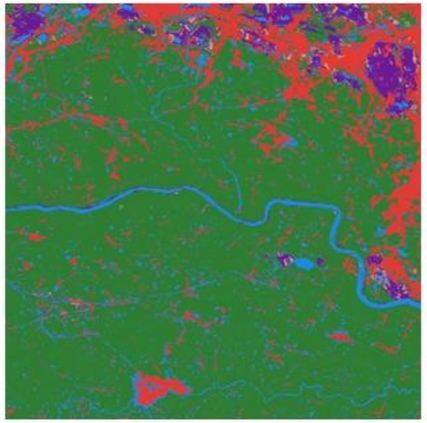


2 Alpha Earth: 64 numbers per 10 m pixel, summing up its year

3 of the 64 shown as colour



3 Base model as band math: a base class for every pixel



4 Walk your class tree; a split only touches its parent's pixels

here barren into mining (purple) and the rest

5 Out: map tiles on screen, a GeoTIFF on request

WHY IT'S BUILT THIS WAY

Linear models replay as plain band math: nothing is downloaded, no GPU needed

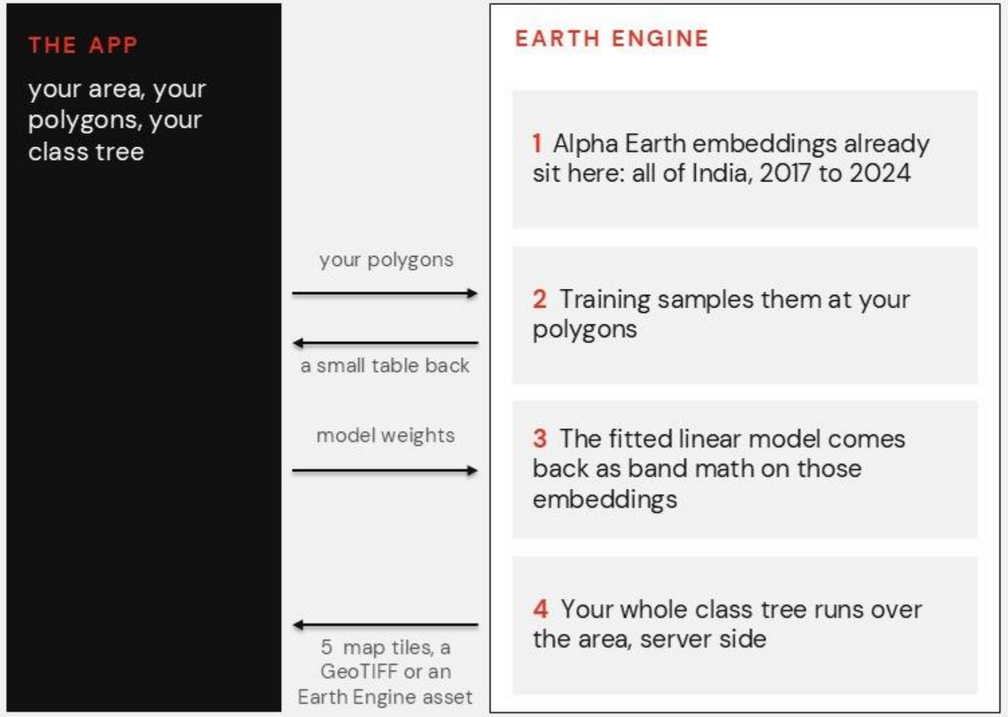
A new class trains from a few of your polygons, inside Earth Engine

Each class you add is its own small model; the shared base map underneath stays as it is

Every run is saved with the exact scheme that made it

Methodology: how we run it

OUR PIPELINE ON EARTH ENGINE



WHAT THIS GIVES US

No imagery is downloaded and no GPU is needed anywhere

Any area of India at 10 m, for any year Alpha Earth covers

The app itself is one small container: front page, tool and API

Every run is saved with the scheme and weights that made it, plus a STAC record

THE SECOND EMBEDDING: TESSERA

Splits can also be trained on Tessera embeddings: 128 numbers per 10 m pixel, for finer detail

Tessera is sampled outside Earth Engine, so it also takes non-linear models such as Random Forest

The pipeline: one run, end to end

BROWSER → APP CONTAINER → AIRFLOW DAG → EARTH ENGINE → MAP + STAC RECORD

1 YOU ASK

You press Run, Train or Download

The app checks the project is yours, then triggers the DAG

2 IT COMPUTES

Airflow calls the app back with the area, year and project

Earth Engine walks your class tree over Alpha Earth

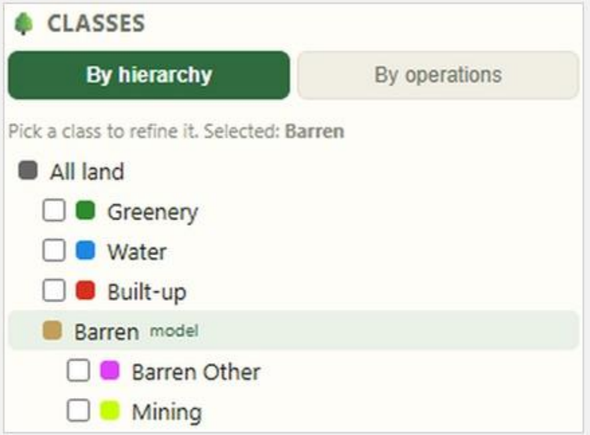
3 IT LANDS

Map tiles on screen, a GeoTIFF on request

STACD registers the STAC record; the run is saved with its scheme

Your classes grow as a tree

HOW THE TREE WORKS



1 Start from base classes: a base scheme such as IndiaSAT's or WorldCover's

2 Pick any class and split it into finer ones, at any depth

3 Each split only decides among its parent's pixels

4 e.g. Barren into mining and the rest (shown); Greenery into acacia and non acacia

5 The tree is saved with every run, so any map can be rebuilt

WAYS TO CHANGE A CLASS

ZOO MODEL reuse one someone trained; rename its classes

YOUR DATA upload labelled polygons, then train

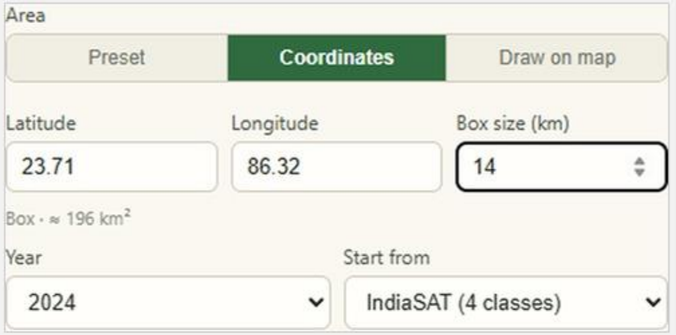
RULE an index and a threshold (NDVI above 0.3), no training

ADD pull a new class out of an existing one

MERGE classes from different splits into one

What the app accepts

AREA AND YEAR



Anywhere in India: a preset, typed coordinates with a box size, or a box drawn on the map

Any year the embeddings cover, 2017 to 2024; the year picker shows which

The area is fixed once the project has a run; a new area is a new project

Very large areas: the map works, and the app tells you if a GeoTIFF export is too big for that area

YOUR POLYGONS: THE UPLOAD FORMAT

```
{ "type": "FeatureCollection",
  "features": [ {
    "type": "Feature",
    "properties": { "class": "acacia" },
    "geometry": { "type": "Polygon", "coordinates": [ ... ] }
  } ] }
```

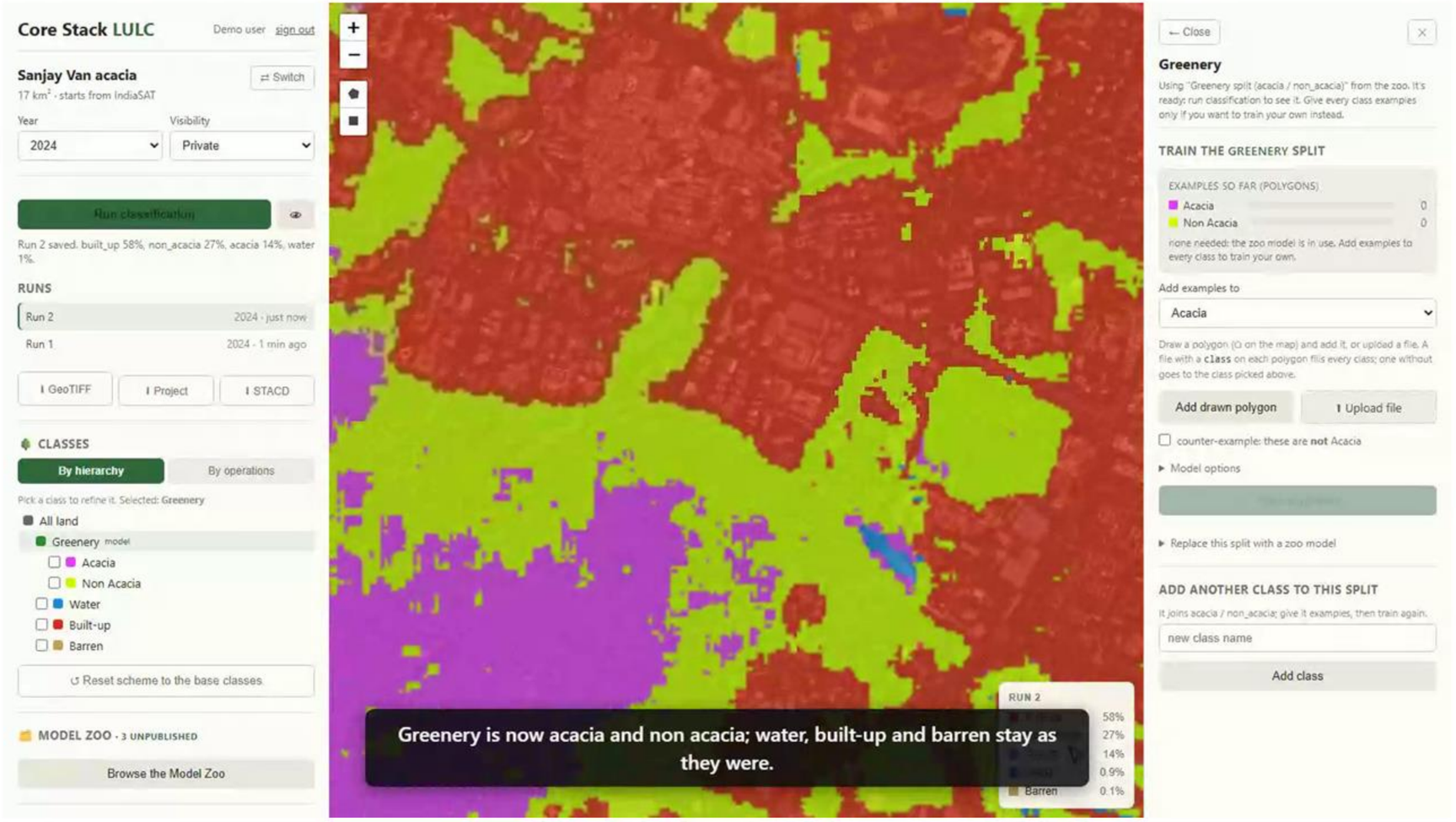
One GeoJSON FeatureCollection in EPSG:4326; polygons only, each with a class

Optional role: negative marks a counter example

The classes are read off the file; one upload fills a whole split

An example file sits in the upload box; a bad file is refused with the exact reason

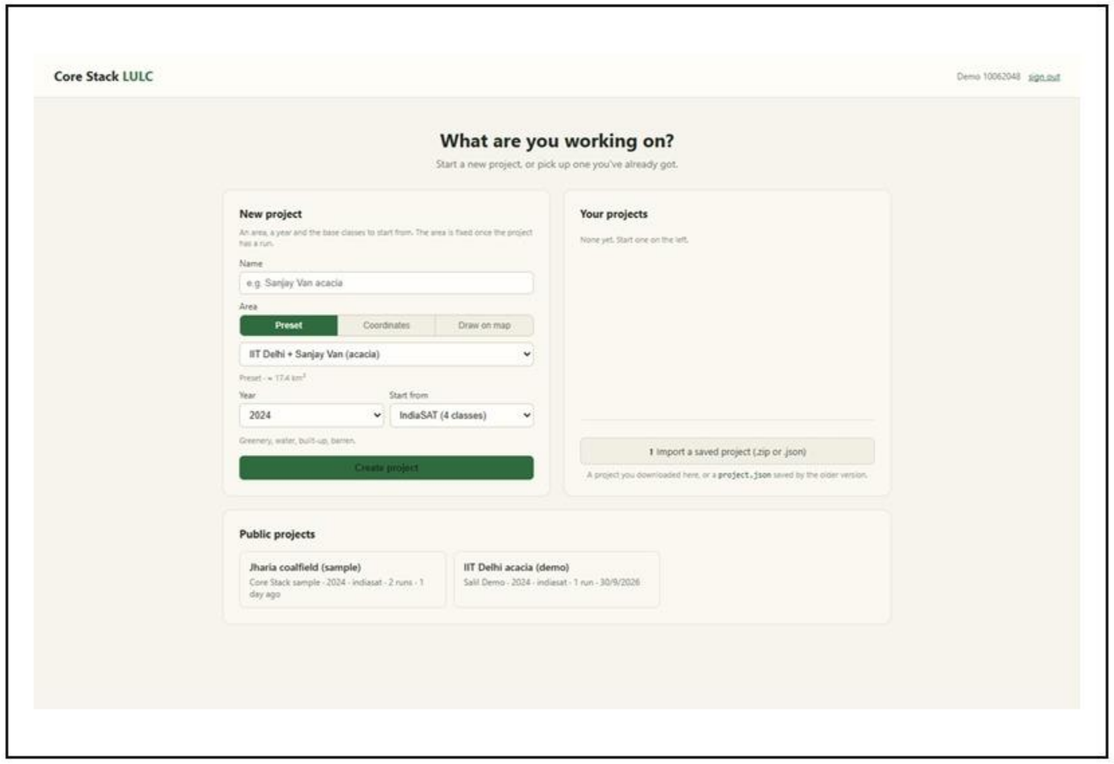
Watch: acacia in Sanjay Van



CHAPTERS · 2:46

0:03 DIY LULC
0:07 Load a base classifier
0:43 Split a class
1:22 Acacia across Sanjay Van
2:16 Save the output
2:38 That's it

Start a project

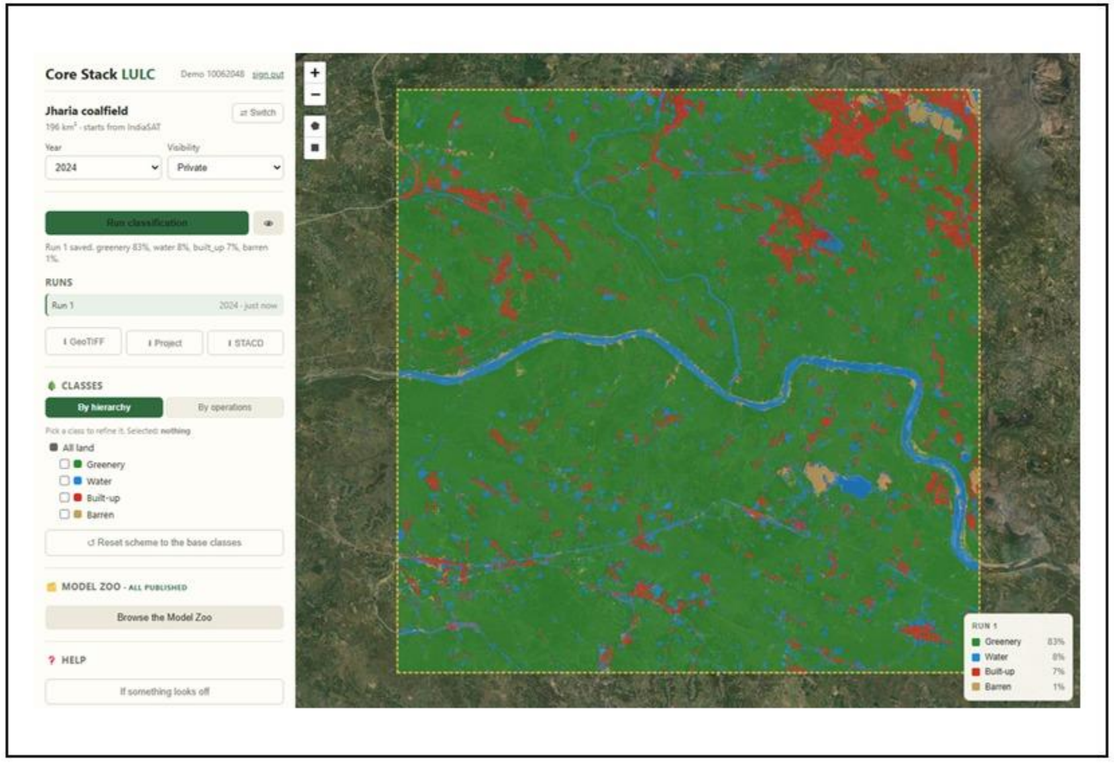


1 Sign in with Google. You only ever see your own projects.

2 New project: a name, an area (preset, coordinates or a box), a year and the base classes.

3 Or pick up an older one from Your projects, or open anyone's public project read only.

Your first map

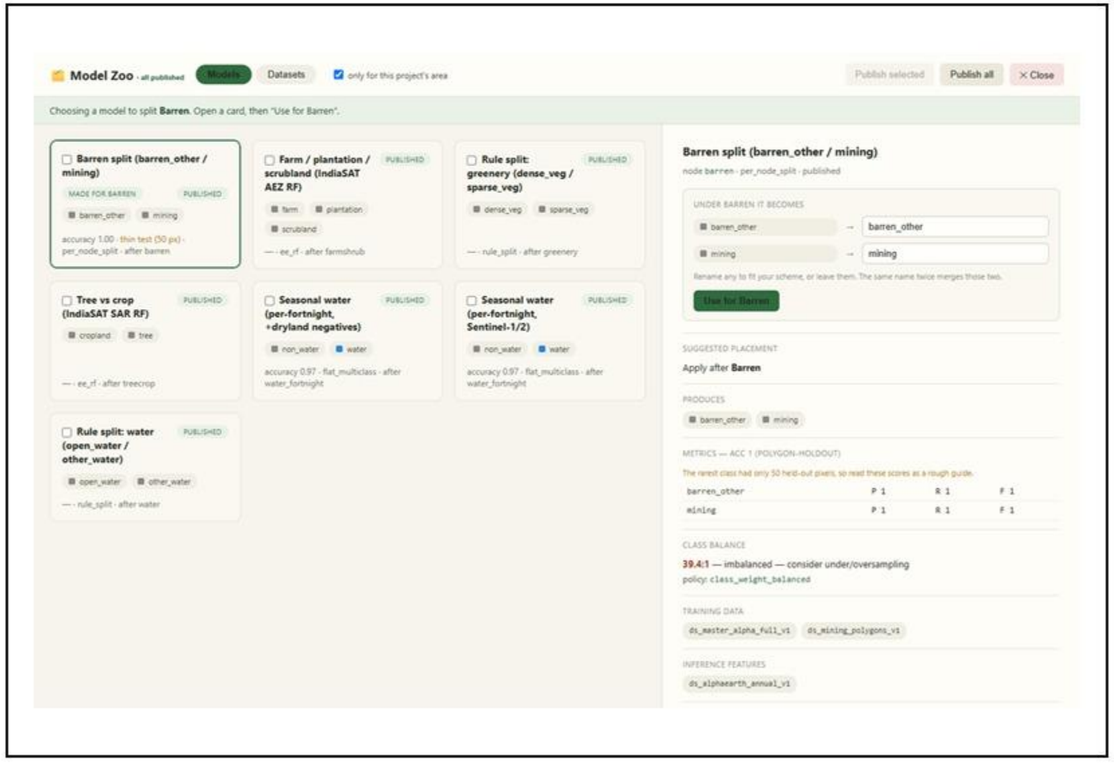


1 Run classification draws the base map over your area: greenery, water, built-up and barren.

2 Class shares show under the button, and a clock shows it's working.

3 It is saved as Run 1, with the exact scheme that made it.

Split a class with a zoo model

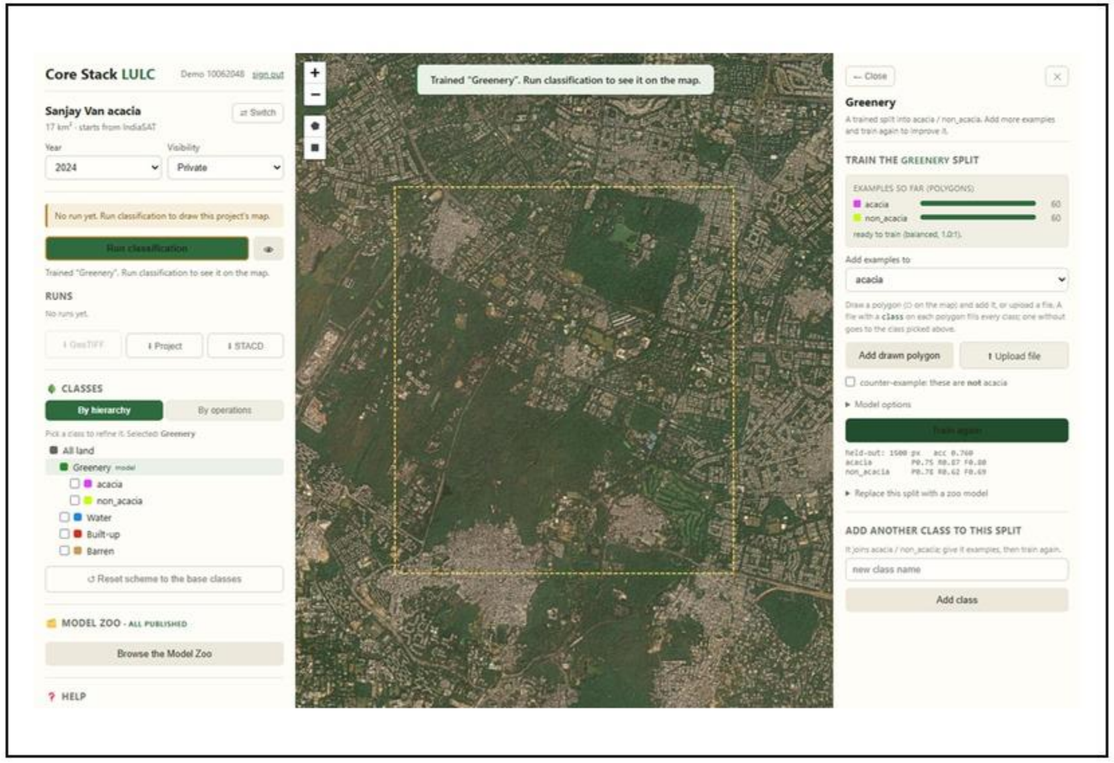


1 Pick a class; the panel shows only what fits it. Choose a model from the zoo.

2 On its card, keep each class name or type your own. The same name twice merges two.

3 Use it: the panel says the model is ready. Run classification to see it.

Split a class with your own data

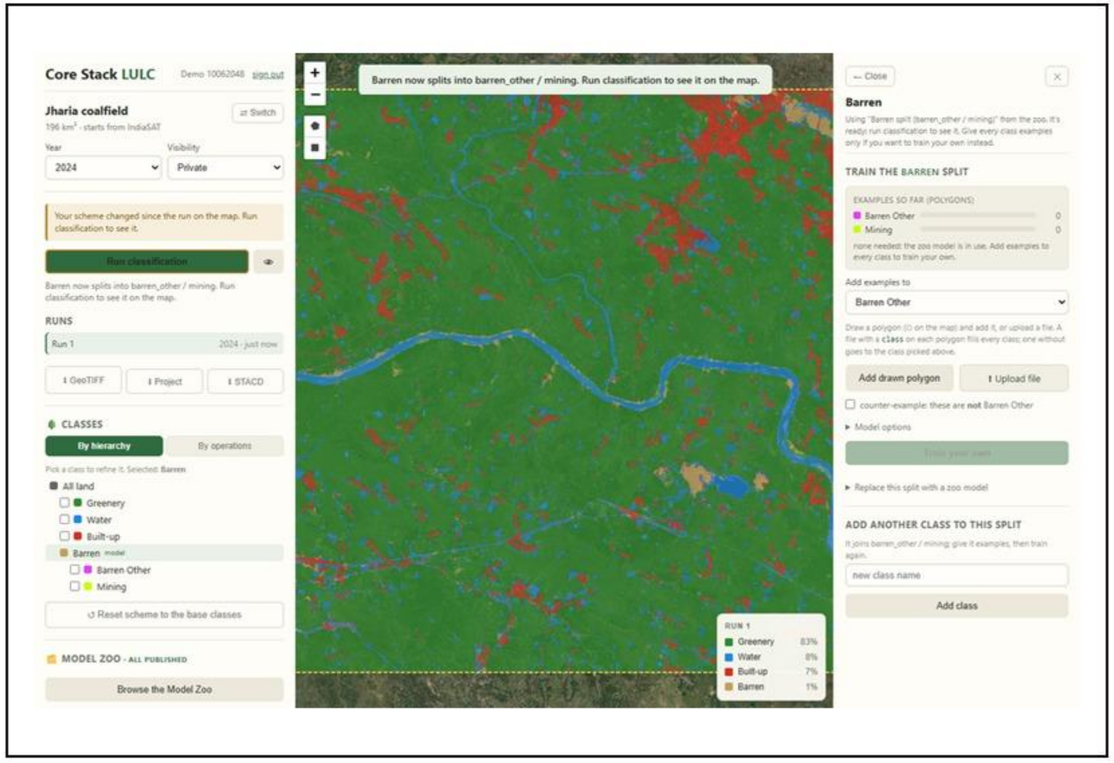


1 Build your own: upload labelled polygons. The split is made from the classes in the file.

2 A bar per class counts the examples and flags lopsided ones.

3 Train: precision, recall and F1 per class on pixels the model never saw. Low recall, that class gets missed.

Runs: nothing runs by itself

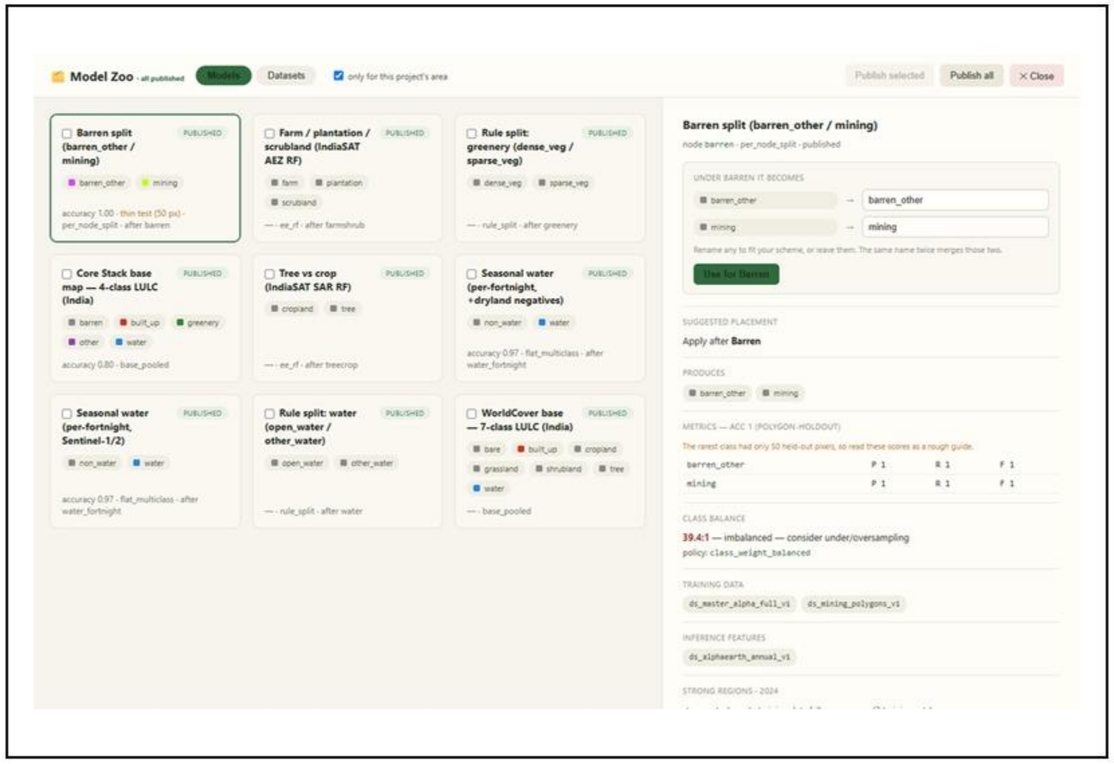


1 Any change raises a banner: the scheme changed since the run on the map.

2 Run when you're ready. Every run is kept; click an older one to put its map back.

3 Open the project another day and the last run is drawn straight away.

Share it, and the model zoo



1 Visibility: Public lists the project on the front page.

2 Visitors see it read only and can copy it into their own projects.

3 Every shared model has a zoo card: what it splits, its scores, its data and where it works well.

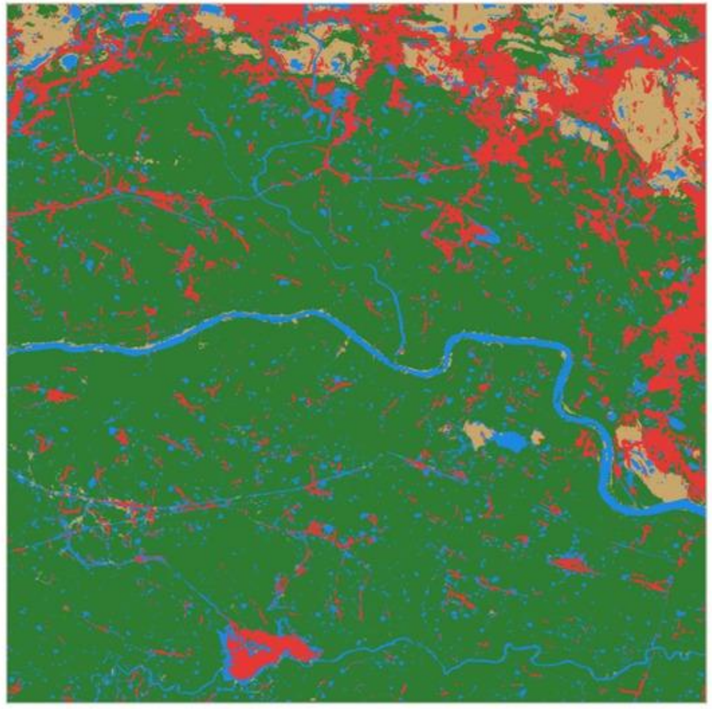
Before and after a split

Jharia coalfield · 18 × 20 km (365 km²) at 10 m · 3.7 million pixels, classified inside Earth Engine

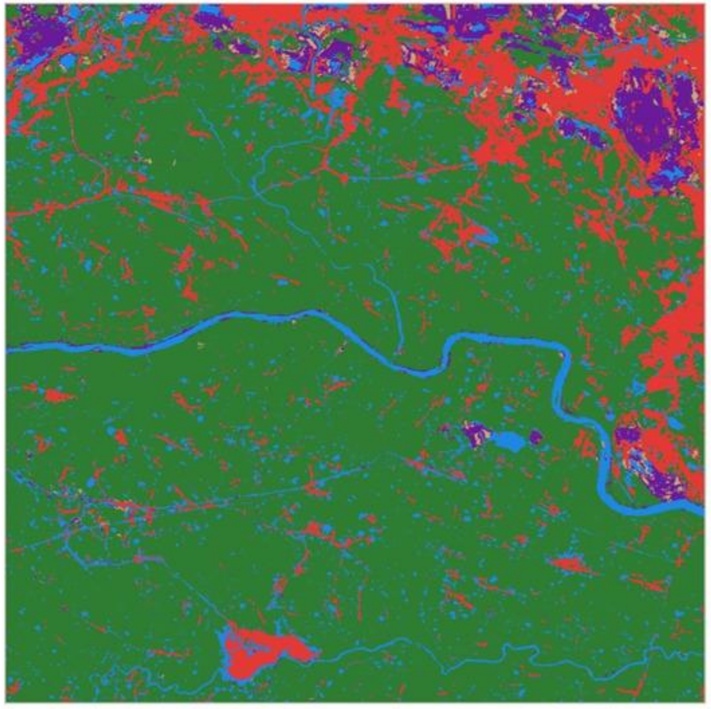
What the user sees
(Sentinel-2, Jan–Apr 2024)



Run classification
4 base classes



After splitting barren
into mining + other



Greenery Built-up Water Barren Mining

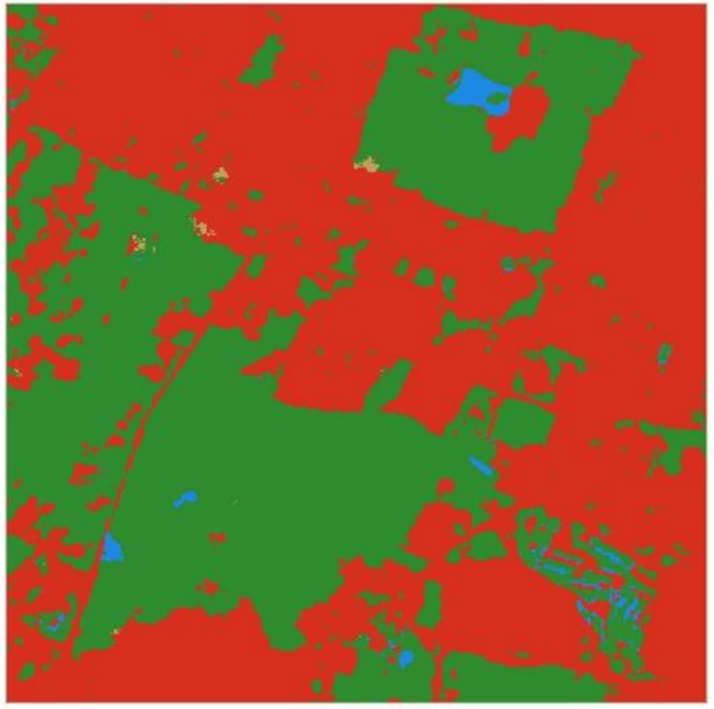
Acacia in Sanjay Van

IIT Delhi + Sanjay Van · 3.9 × 4.4 km (17 km²) at 10 m · acacia 2.4 km², non acacia 4.8 km²

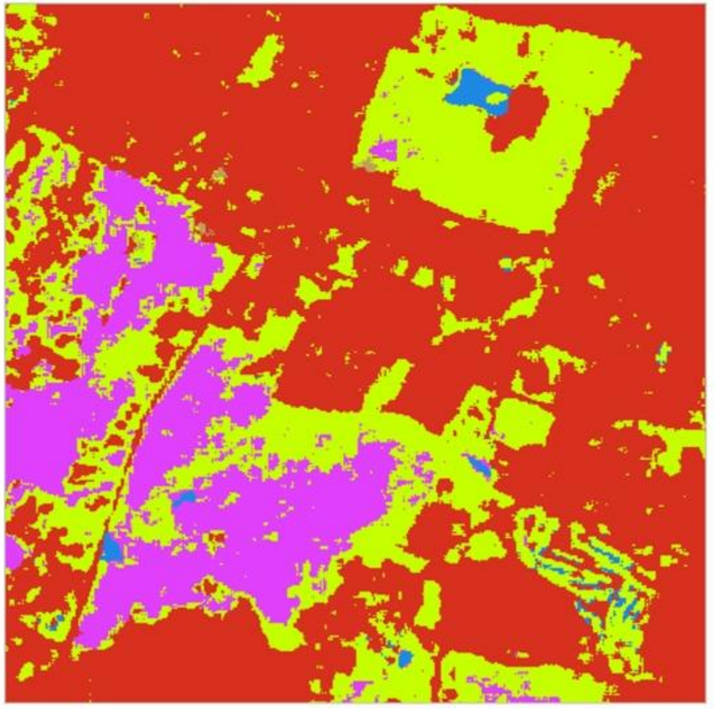
What the user sees
(Sentinel-2, Jan–Apr 2024)



Run 1: the base classifier
4 classes



Run 2: greenery split with the drone-trained acacia model



Acacia · Non acacia · Greenery · Built-up · Water · Barren

What you get

EVERY RUN IS KEPT · with the scheme that made it · reopen it any day · nothing reruns by itself

GEOTIFF

One band of class codes at 10 m, with the code to class legend

The app says if an area is too big to export as one file

PROJECT ZIP

Scheme, your examples, your models and every run

Import it back any time, here or on another server

STAC RECORD

What was made and how: area, year, class tree, steps

So the map can be catalogued and found later

Inside your download

THE PROJECT ZIP

project.json · name, area, year, base map

hierarchy.json · your class tree

examples/ · the polygons for each class

weights/ · models you trained or applied

op_log.json · every change, in order

runs/ · one folder per run, scheme frozen

WHAT TO DO WITH IT

Open the GeoTIFF in QGIS; the class legend is in run.json

The STAC record says what was made and how

Import the zip on the start screen to pick up where you left off

If something looks off

IF YOU SEE	TRY THIS
A class painted where it isn't	Split it, or add counter examples
A split class mostly missed	Add polygons spread across the area, train again
Your upload is refused	Read the reason; start from the example file
The map didn't change after an edit	Press Run classification; nothing reruns by itself
Download says the area is too big	Make a smaller project for that patch
A zoo model fits your area badly	Check its card, or train your own

Thank you

By Salil Gujar · Custom LULC